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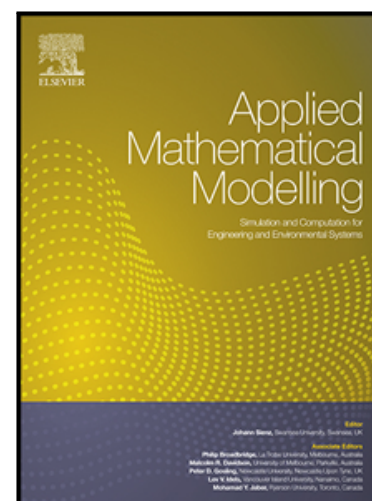
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Highlights

- We apply persistent homology methods for quantitative analysis of the rock pore space changes during chemical dissolution.
- Use the bottleneck metric in the persistence diagrams space allows to classify dissolution scenarios.
- Persistence diagrams allow to estimate the dissolution rate.

Computational topology-based characterization of pore space changes due to chemical dissolution of rocks

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Abstract

In this paper, we present an algorithm for the numerical simulation of reactive transport at the pore scale to facilitate observation of pore space and rock matrix evolution. Moreover, simulation at the pore scale opens up the possibility of estimating changes in the transport properties of rocks, such as permeability and tortuosity. To quantitatively analyze pore space evolution, we developed a numerical algorithm that can be used to construct persistence diagrams of the connectivity components for pore space and the rock matrix, which characterize the topology evolution during rock matrix dissolution. Introducing the bottleneck metric in the space of the persistence diagrams, we cluster the numerical experiments in terms of similarities in topology evolution. We demonstrate that the application of this metric to the persistence diagrams allowed us to distinguish topologically different dissolution scenarios, for instance, the formation of a dissolution front near the inlet, homogeneous dissolution of the matrix inside the core sample, and formation of wormholes. We illustrate that the differences in topology evolution lead to cross-correlations among the transport properties of rocks (porosity-permeability-tortuosity).

Keywords: rock dissolution, finite-differences, level-set, computational

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topology, Betti numbers, persistence diagram

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1. Introduction

Numerical simulation of reactive transport has numerous applications, including enhanced oil recovery through acidization and acid hydro-fracturing [1], CO_2 sequestration in carbonate formations [2], bio-cementation [3], salt precipitation [4], and injection of non-condensable gases into geothermal fields [5]. In the abovementioned applications, heterogeneous reactions, which are the chemical fluid-solid interactions occurring at the interface between pore space and rock matrix, constitute the target of interest. The result of these interactions is matrix dissolution (acidization) or mineral precipitation (bio-cementation). Chemically induced micro-scale changes in pore space or the rock matrix affects the macro-scale physical properties of the rocks. Specifically, rock matrix dissolution increases the porosity, permeability, and mechanical compliance of rock masses [6]. Moreover, changes in the specific surface area and tortuosity may affect the reaction rate at the macro scale. Predictive reactive transport simulations are mainly performed at the reservoir scale [7]. In such models, porosity satisfies the evolutionary equation, where the rate of the porosity changes depends on the reaction rate and concentration of active species inside a grid cell, and the reaction rate, permeability, tortuosity, and diffusion coefficient are related to porosity according to the empirical equations of state [7]. These relationships can be obtained either from laboratory measurements [6] or by using averaging techniques [8]. However, only a few laboratory experiments have been reported in the literature because of their time-intensive nature. Moreover, laboratory experiments can be conducted only once with a given sample. By contrast, averaging has been applied to relatively simple models, typically having periodic structures, which is not the case with rocks. Thus, numerical simulation of chemical fluid-solid interactions at the pore scale may help us construct the relationships among the main macroscopic parameters under different

conditions of flow, inlet species concentrations, and reaction rates [9]. According to previous laboratory experiments [6], and numerical simulations conducted at the pore scale [10] changes in the pore space geometry and topology of a rock depend on the chemical reaction rate in the pore space, fluid flow rate, molecular diffusion, and other parameters defined at the pore scale. Furthermore, changes in the geometry govern the differences in the effective parameters at the macro- or reservoir scale, which are used in reactive transport simulators, such as TOUGHREACT, PFLOTRAN, and PHREEQC; see [7] for a review.

Numerical simulation of reactive transport at the pore scale involves directly computing the fluid flow at the pore scale, solving the convection-diffusion equation to model the active component transport, and simulating the heterogeneous reaction occurring at the pore space-rock matrix interface [9]. The principal difference among the existing approaches is the manner of approximation of the boundary conditions at the moving interface. There are three main types of methods. Methods of the first type employ regular rectangular meshes to approximate all equations (Stokes, Navier-Stokes, convection-diffusion), where the term effective relative mass of fluid or fluid "concentration" is introduced for the "boundary" grid cells. The relative mass changes according to additional law that is related to the true reaction rate [11], [12]. These methods are easy to implement, but the law defining the rate of relative changes in mass is mainly postulated, that is, it is empirical. The method of the second type, namely the front-tracing method, is based on explicit representation of the interface. This approach is usually combined with the finite-volumes with truncated cells approach [9]. However, this method is difficult to implement when the topology and geometry of the domain (pore space) are complex, especially if the topology of the domain changes. Methods of the third type are based on implicit representations of the sharp interface, where the exact boundary conditions are stated. These include the level-set [13], and phase-field [14] methods. These methods allow one to use the original chemical kinetics relationships at the pore scale, and they are relatively simple to implement, even in case where the pore space topology changes. Moreover, in these methods, regular rectangular

grids are used to approximate the equations inside pore space, whereas boundary conditions are approximated using the immersed boundary method [15], [16].

This paper presents numerical algorithms for constructing persistence diagrams that allow one not only to evaluate the topology of a pore space but also to present compressed information about changes to the pore space and the related matrix topology. In particular, the persistence diagrams contain information about each connectivity component (isolated pore) or cycle and the times at which these components appear and disappear. Thus, these diagrams can be used to estimate the rate of topology changes and measure dissimilarities in the evolution of pore spaces. Usually, persistence diagrams are computed for simplicial complexes (Chapter VII in [17]), but in this study, we apply persistence diagrams to digital images [18]. In doing so, the main challenge lies in defining the type of point (voxel) connectivity that can ensure both consistency and topological duality. To find the connected components at each level of filtration and construct a persistence diagram, we employ a disjoint-set data structure. To estimate the proximity of pore spaces topologies, we use the "bottleneck" metric in the space of persistence diagrams; this metric is based on the selection of the best one-to-one correspondence between diagrams. Other metrics, such as the Hausdorff metric, have several disadvantages. In particular, the Hausdorff metric does not consider (or poorly considers) the significant weights of clusters of points on persistence diagrams. However, these clusters reflect the essential features of the changes in pore space topology. There is now an intrinsic vector structure in the space of all persistence diagrams; therefore, the problem of averaging over a collection of diagrams is nontrivial. One way is to find the mean in the Fréchet sense. The corresponding algorithms based on the p -th Wasserstein metric were developed in [19]. We use the bottleneck metric to cluster the simulation results based on similarities in pore space topology changes under various experimental conditions, such as flow rate, molecular diffusion coefficient, and reaction rate. We apply the algorithms to investigate the influence of the microscopic parameters, such as flow rate, molecular diffu-

sion coefficient, and reaction rate, on the changes in pore space. Moreover, we show that the bottleneck metric allows us to cluster the persistence diagrams corresponding to different topology evolutions.

The remainder of this paper is organized as follows. In section 2, we present the algorithm for the pore-scale numerical simulation of reactive transport. We define persistence diagrams and the bottleneck metric in section 3. In section 4, we describe the numerical experiments conducted to illustrate the applicability of persistence diagrams to the classification of pore space evolution as a function of the reactive flow parameters.

2. Reactive flow simulation

2.1. Problem Statement

In this paper, we assume that the presence of chemically active components dissolved in a fluid does not affect the physical properties of the fluid. Thus, a single-phase fluid flow can be considered. Moreover, we assume that the flow rate is low, that is, the Reynolds number is considerably smaller than one, and the flow attains a steady state at a faster pace than the rate at which geometry changes. Thus, the flow satisfies the steady-state Stokes equations.

Consider a restricted domain $D \subseteq R^2$ composed of two multiconnected non-overlapping time-dependent subdomains $D_p(t)$ and $D_m(t)$ representing the pore space and the matrix, respectively. The outer boundary of this domain is represented by $\partial D = \Gamma_{outlet} \cup \Gamma_{inlet} \cup \Gamma_{nf}$, where Γ_{outlet} is the outlet boundary, Γ_{inlet} the inlet boundary (opposite to the outlet), and Γ_{nf} the no-flow part of the boundary. Moreover, we introduce the interface between the pore space and the matrix $\bar{D}_p(t) \cap \bar{D}_m(t) = \Gamma(t)$. We assume that $\Gamma(t)$ is composed of sufficiently smooth curves. A sketch of the computational domain is shown in Figure 1.

The fluid flow is defined in the domain $D_p(t)$ and it satisfies the steady-state

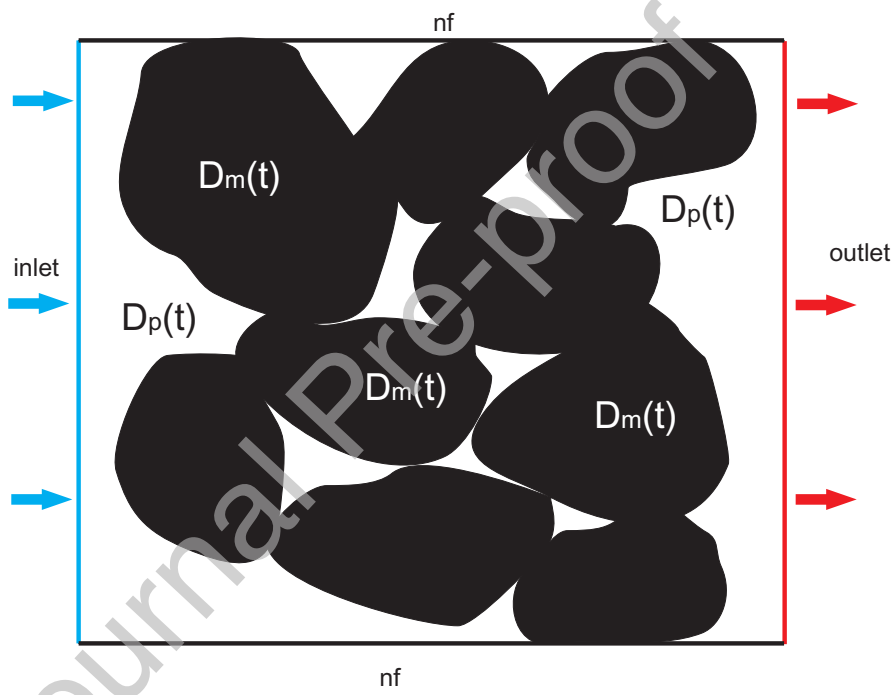


Figure 1: Sketch of the computational domain.

Stokes equation:

$$\begin{aligned}\mu \nabla^2 \vec{u} - \nabla p &= 0, \\ \nabla \cdot \vec{u} &= 0,\end{aligned}\tag{1}$$

with the boundary conditions

$$\begin{aligned}\vec{u}(\vec{x}) &= 0, & \vec{x} \in \Gamma(t) \cup \Gamma_{nf}, \\ p(\vec{x}) &= p_{bc}(\vec{x}), & \vec{x} \in \Gamma_{inlet} \cup \Gamma_{outlet},\end{aligned}\tag{2}$$

where $\vec{u} = (u_1, u_2)^T \in R^2$ is the velocity vector, p the pressure, μ the dynamic viscosity coefficient, $p_{bc}(\vec{x})$ the pressure at the boundary, and $\vec{x} = (x_1, x_2)^T$ the vector of spatial coordinates. For simplicity, we impose the no-flow boundary condition at the top and bottom boundaries, which allows us to estimate only the horizontal component of the permeability tensor if the sample size exceeds the representative volume [20]. To estimate the full permeability tensor, we may need to impose advanced boundary conditions, for example, periodic boundary conditions; see [21], [22] for discussion.

Transmission of the reactive component inside the domain $D_p(t)$ is described using the convection-diffusion equation

$$\frac{\partial C}{\partial t} + \nabla \cdot (\vec{u}C - D\nabla C) = 0,\tag{3}$$

with the boundary conditions

$$\begin{aligned}D \frac{\partial C}{\partial \vec{n}} &= k_r(C - C_s), & \vec{x} \in \Gamma(t), \\ C &= C_{in}, & \vec{x} \in \Gamma_{inlet}, \\ \frac{\partial C}{\partial \vec{n}} &= 0, & \vec{x} \in \Gamma_{outlet} \cup \Gamma_{nf}.\end{aligned}\tag{4}$$

Here D is the diffusion coefficient, C the molar concentration of the active component, $\vec{n}$ the vector normal to the boundary or interface and directed inside the domain D_p , and k_r the coefficient that determines the reaction rate. Note, that we consider only first-order reactions, in which the reaction rate linearly depends on the concentration of the active component. Moreover, we consider that the rock matrix is monomineralic and the solution contains only one active component; thus, the right-hand side of equation (3) equals zero.

The evolution of the interface $\Gamma(t)$ satisfies

$$v_n(\vec{x}, t) = \frac{K_c k}{\rho}(C - C_s), \quad \vec{x} \in \Gamma(t), \quad (5)$$

where v_n is the normal component of the boundary changing velocity, K_c the stoichiometric reaction coefficient, ρ the mass density of the matrix, and C_s the equilibrium concentration. Note that the value of C_s depends on the reaction. Moreover, expression $C - C_s$ is positive for dissolution and negative for precipitation. In this study, we consider cases of dissolution, that is, $C - C_s > 0$.

2.2. Level-set method

In this section, we briefly review the level-set method to implicitly define the interface $\Gamma(t)$. In principle, this method considers the interface as the constant value of the scalar function $\varphi(\vec{x}, t)$:

$$\Gamma(t) = \{\vec{x} | \varphi(\vec{x}) = 0\}.$$

Zero is a natural choice for the goal level. In this case, the domains D_p and D_m are defined as

$$D_p(\vec{x}, t) = \{\vec{x} | \varphi(x, t) > 0\}, \quad D_m(\vec{x}, t) = \{\vec{x} | \varphi(x, t) < 0\}.$$

Typically the level-set function is the signed distance, that is, $\|\nabla_x \varphi(\vec{x}, t)\| = 1$. In this case, the vector normal to the interface is

$$\vec{n} = \nabla_x \varphi(\vec{x}, t)|_{\Gamma(t)}.$$

The change in interface geometry, as determined using the level-set function, is ([23])

$$\begin{aligned} \frac{\partial \varphi(\vec{x}, t)}{\partial t} + v_n(\vec{x}, t) &= 0, \\ \varphi(\vec{x}, 0) &= \varphi_0, \end{aligned} \quad (6)$$

where v_n is the normal component of interface velocity, as defined in equation (5).

Note that the slowest time scale in reactive transport is the one in equation (6); this time scale defines the overall time scale of the problem.

2.3. Numerical schemes

We use the same rectangular grid to solve all the considered equations, namely Stokes equation (1), convection-diffusion equation (3), and level-set evolution equation (6). We use the staggered-grid finite-difference method to approximate the equations. The grid functions are defined as $p_{i,j} = p(ih_1, jh_2)$, $(u_1)_{i+1/2,j} = u_1((i + 1/2)h_1, jh_2)$, $(u_2)_{i,j+1/2} = u_2(ih_1, (j + 1/2)h_2)$, $C_{i,j}^n = C(ih_1, jh_2, n\tau)$, and $\varphi_{i,j}^n = \varphi(ih_1, jh_2, n\tau)$, where h_1 and h_2 are the spatial grid steps, and τ is the temporal grid step (either physical or model time parameter is considered here). If a level-set function is defined at the centers of the grid cells, the interface is piecewise linear. Note that domain discretization is defined by the model, which is usually developed based on computed tomography scans of rocks [24] and is adequately fine, such that it can yield accurate numerical solutions with low-order approximations.

2.3.1. Stokes equation

To approximate Stokes equation, we apply the second-order staggered-grids finite difference scheme:

$$\begin{aligned} \mu D_1^2[u_1]_{i+1/2,j} + D_2^2[u_1]_{i+1/2,j} - D_1^c[p]_{i+1/2,j} &= 0, & \text{if } \varphi_{i+1/2,j} > 0, \\ \mu D_1^2[u_2]_{i,j+1/2} + D_2^2[u_2]_{i,j+1/2} - D_2^c[p]_{i,j+1/2} &= 0, & \text{if } \varphi_{i,j+1/2} > 0, \\ D_1^c[u_1]_{i,j} + D_2^c[u_2]_{i,j} &= 0, & \text{if } \varphi_{i,j} > 0, \end{aligned} \quad (7)$$

where D_1^c and D_2^c are the central differences with respect to the spatial directions x_1 and x_2 , respectively, and D_1^2 and D_2^2 are the operators for approximating the second-order derivatives:

$$D_1^2[f]_{I,J} = \frac{f_{I+1,J} - 2f_{I,J} + f_{I-1,J}}{h_1^2} = \frac{\partial^2 f}{\partial x_1^2} \Big|_{I,J} + O(h_1^2), \quad (8)$$

$$D_1^c[f]_{I,J} = \frac{f_{I+1/2,J} - f_{I-1/2,J}}{h_1} = \frac{\partial f}{\partial x_1} \Big|_{I,J} + O(h_1^2). \quad (9)$$

Here, the indices denoted by the uppercase letters I and J can be either integer or half-integer, whereas the indices denoted by the lowercase letters i and j can only be integer. To construct the operators approximating the derivatives

with respect to other spatial directions, the roles of the spatial indices should be permuted.

Equation (7) is defined at the grid points with $\varphi_{I,J} > 0$, but a few of the points involved in the stencil may belong to domain D_m . In this case, we suggest using the Immersed Boundaries Methods (IBM) [25]. This method extrapolates the solution to the domain D_m over the interface $\Gamma(t)$ by considering the original boundary conditions. Assume the point $((x_1)_{\hat{I}}, (x_2)_{\hat{J}}) \in D_m(t)$; a solution is needed at this point. With the no-slip boundary condition at the interface, $\vec{u}_{\Gamma(t)} = 0$. In this case, the velocity components through the interface can be thought of as an odd function, whereas pressure can be considered an even function. To construct the extrapolation, the orthogonal reflection of this point over the interface should be constructed. The vector normal to the interface can be estimated as $\vec{n} = \nabla_x \varphi(\vec{x})|_{\hat{I}, \hat{J}}$, and the distance to the interface is the value of the level-set function. Thus, the coordinates of the point at the interface at which the normal line passing through $((x_1)_{\hat{I}}, (x_2)_{\hat{J}})$ intersects the boundary are

$$((x_1)_c, (x_2)_c) = ((x_1)_{\hat{I}}, (x_2)_{\hat{J}}) + \varphi_{\hat{I}, \hat{J}} \vec{n},$$

and the coordinates of the reflected point are

$$((x_1)_n, (x_2)_n) = ((x_1)_{\hat{I}}, (x_2)_{\hat{J}}) + 2\varphi_{\hat{I}, \hat{J}} \vec{n}.$$

Moreover, the following relationship holds:

$$\vec{u}_{\hat{I}, \hat{J}} = -\vec{u}((x_1)_n, (x_2)_n) + O(\varphi_{\hat{I}, \hat{J}}^2).$$

Then, the solution at the reflected point $((x_1)_n, (x_2)_n)$ can be interpolated. The standard approach involves performing bilinear interpolation based on the values of four neighboring points. However, if rock pore space is considered, and especially, if changes to the topology of the pore space or the matrix are considered, we commonly end up with a few neighboring points belonging to the matrix, that is, to the domain D_m , where the solution is not defined. To overcome this problem, we suggest the use of linear interpolation based on three values defined for three points that are the closest to point $((x_1)_n, (x_2)_n)$.

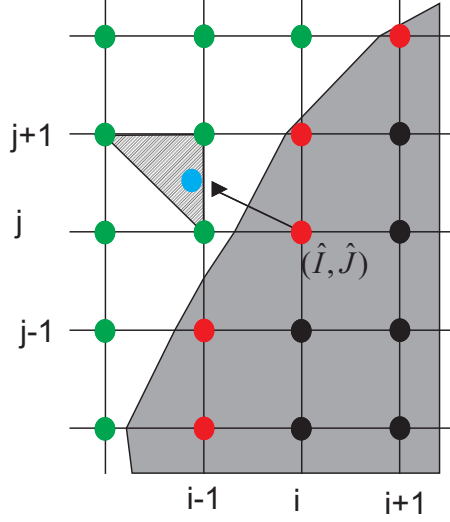


Figure 2: A sketch of solution extrapolation. The grey domain is the rock matrix or domain $D_m(t)$, and the white area is the pore space or domain Ω . The black dots are not involved in the computations, the solutions at the red points are extrapolated, and the solutions at the green points are computed using finite differences. The blue point is a reflected point with the coordinates $((x_1)_n, (x_2)_n)$. The dots at the vertices of the filled triangle are used for interpolation.

Moreover, if the Dirichlet condition is considered, which is the case for the velocity components, we additionally use point $((x_1)_c, (x_2)_c)$ at the boundary, where the function is known owing to the boundary condition. A sketch of the IBM used for the Dirichlet condition is shown in figure 3.

To extrapolate pressure, we use the corollary of the original boundary conditions $\frac{dp}{dn} = 0$. As a result, the pressure at the immersed point $((x_1)_j, (x_2)_j)$ can be expressed as:

$$p_{\hat{i}, \hat{j}} = p((x_1)_n, (x_2)_n) + O(\varphi_{\hat{i}, \hat{j}}^2).$$

Thereafter, the solution at the reflected point $((x_1)_n, (x_2)_n)$ can be interpolated using the values at the three nearest points, which may include the immersed point $((x_1)_j, (x_2)_j)$, as in figure 3.

We described the algorithm to approximate the Stokes equation in domains

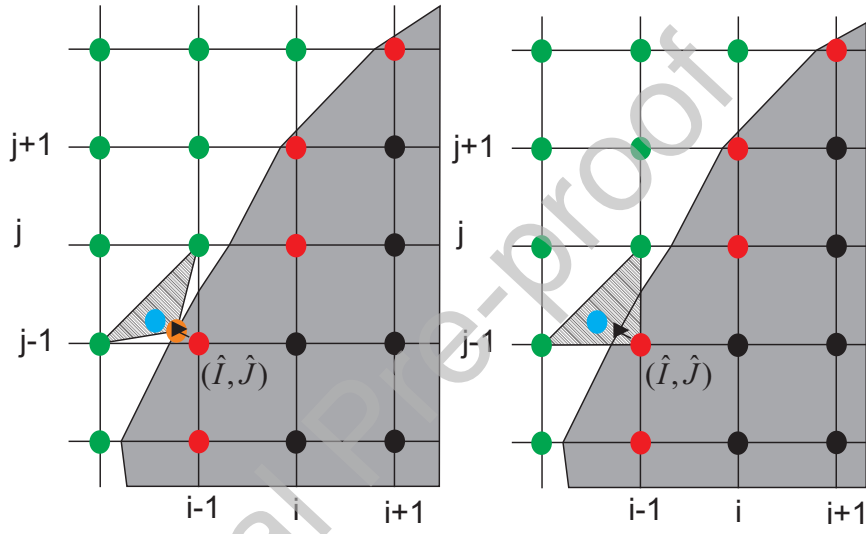


Figure 3: A sketch of solution extrapolation for the Dirichlet (left) and Neumann (right) boundary conditions when the immersed point is close to the interface. The grey domain is the rock matrix or the domain $D_m(t)$, and the white area is the pore space $D_p(t)$. The black dots are not involved in the computations, the solution at the red points is extrapolated, and the solution at the green points is computed using finite differences. The blue point is the reflected point with the coordinates $((x_1)_n, (x_2)_n)$. The dots at the vertices of the filled triangle are used for interpolation.

with complex geometry. In doing so, we defined the action of the matrix at the vector of the grid functions. Thus, any appropriate iterative solver can be used to solve the derived system of linear algebraic equations.

2.3.2. Advection-diffusion equation

For the sake of completeness we provide the first-order accurate up-wind finite-difference approximation of the convection-diffusion equation used in our research:

$$\frac{C_{i,j}^{n+1} - C_{i,j}^n}{\tau} + D_1^1[u_1 C]_{i,j}^n + D_2^1[u_2 C]_{i,j}^n - DD_1^2[C]_{i,j}^n - DD_2^2[C]_{i,j}^n = 0, \quad (10)$$

$$\varphi_{i,j} > 0,$$

where

$$D_1^1[u_1 C]_{i,j}^n = \frac{F_{i+1/2,j} - F_{i-1/2,j}}{h},$$

$$F_{i+1/2} = \begin{cases} (u_1)_{i+1/2,j} C_{i+1}, & (u_1)_{i+1/2,j} < 0 \\ (u_1)_{i+1/2,j} C_i, & (u_1)_{i+1/2,j} > 0 \end{cases} \quad (11)$$

The operators D_1^2 and D_2^2 were introduced in equation (8). Note that we use low-order approximation for two reasons. First, the domain discretization used in digital rock physics is extremely fine. As a result, the effects of numerical dissipation are relatively weak, and the solution is adequately accurate. Second, the complex geometries of pore space do not allow for expansion of the finite-difference stencil, thus increasing the complexity of IBM without improving its accuracy.

As in case of the Stokes equation, the main difficulty lies in the approximation of the boundary conditions at the interface $\Gamma(t)$. The concentration of the active component satisfies Robin boundary condition

$$D\nabla C \cdot \vec{n} + k_r C = k_r C_s,$$

which can be considered a linear combination of the Dirichlet and Neumann conditions. As in the previous section, we assume that point $((x_1)_f, (x_2)_f) \in D_m$ belongs to a stencil centered in domain D_p . Thus, we extrapolate the solution at this point. Point $((x_1)_n, (x_2)_n)$ is the orthogonal reflection of the immersed

point over the interface $\Gamma(t)$. The boundary condition can thus be represented as

$$D \frac{C((x_1)_n, (x_2)_n) + C_{\hat{I}, \hat{J}}}{2} + k_r \frac{C((x_1)_n, (x_2)_n) - C_{I, J}}{2\varphi_{\hat{I}, \hat{J}}} = k_r C_s.$$

By resolving this equation with respect to $C_{\hat{I}, \hat{J}}$, we can obtain the extrapolation relationship, where $((x_1)_n, (x_2)_n)$ is interpolated using the three nearest points, as described above.

2.3.3. Interface evolution

To simulate interface evolution, it is sufficient to solve equation (6), which governs the changes in the level-set function

$$\frac{\varphi_{i,j}^{n+1} - \varphi_{i,j}^n}{\tau} = -(v_n)_{i,j}^n, \quad (12)$$

where $(v_n)_{i,j}^n$ is the rate of interface change along the direction normal to the current interface profile.

Equation (6) is defined over the entire domain D . Thus, the right-hand side of this equation is defined in D as well. However, the parameter $v_n(\vec{x})$ is given only at the interface $\Gamma(t)$; thus, it should be extended to the subdomains $D_p(t)$ and $D_m(t)$. Following [13] we extrapolate the rate of evolution of the level-set from the interface inside the domains by a constant value along the direction normal to the level-set, that is, along the gradient. This is achieved by computing the steady-state solution of the following problem:

$$\begin{aligned} \frac{\partial q}{\partial t} + \text{sign}(\varphi) \left(\frac{\nabla \varphi}{|\nabla \varphi|} \cdot \nabla q \right) &= 0, \\ q(\vec{x}, 0) &= \tilde{v}_n(\vec{x}, t_0), \end{aligned} \quad (13)$$

where $\tilde{v}_n$ is the initial condition, which coincides with the normal velocity at the interface and with zero elsewhere.

To approximate the gradient of the level-set function, we use the centered differences, while the derivatives of function $q(\vec{x}, t)$ are approximated using the up-wind scheme [26]. Note, that the steady-state solution of (13) is required in a thin strip near the interface $D_s = \{\vec{x} : |\varphi(\vec{x})| \leq 2\sqrt{h_1^2 + h_2^2}\}$. Thus, in only three to five iterations, we can obtain the steady-state solution inside the strip

D_s . Once the velocity is extrapolated, we can recompute the level-set function with possible correction of the level-set by using re-distancing [13].

3. Persistent homology

In this section, we present necessary information from the persistent homology theory [17], and describe its application in the analysis of pore space. Let X be a connected topological space and $f : X \rightarrow \mathbb{R}$ a continuous function. For $a \in \mathbb{R}$, we define the excursion set $X_a = \{x \in X | f(x) \leq a\}$. We can consider space X as being collected from a continuous family of excursion sets X_a , with a running from $-\infty$ to $+\infty$. Moreover, the topology of the spaces X_a changes only when passing the critical values of a_i , between which there is no change in topology. The persistent homology is applied to calculate the topological complexity of X in terms of the topological complexity of X_{a_i} relative to that of $X_{a_{i-1}}$.

In the classical approach to defining persistent homology, it is usually assumed that X is homeomorphic to the support of some finite simplicial complex K [17]. However, in practical applications, often, X does not have the structure of a simplicial complex but is instead a digital image; in other words, roughly speaking, it is divided into voxels and not into simplexes [18] (pixels, in our case). The main definitions and results can be carried over to our case, and for completeness, we present them below.

By definition, a voxel of dimension n in the Euclidean space $\mathbb{R}^d$ ($n \leq d$) is an n -dimensional cube $Q \subset \mathbb{R}^d$ with a unit edge, vertices located at integer coordinate points, and edges parallel to the coordinate axes. For voxel Q , we can define voxels that are its co-dimensional faces from 1 to n . A digital image of dimension d represents the set $K = \{Q_1, \dots, Q_N\}$ consisting of d -dimensional voxels $Q_1, \dots, Q_N$. To convert the digital space K into the topological space $|K|$, one must specify the type of its continuity. To this end, by fixing some integer k , such that $0 \leq k \leq d - 1$, we declare that the d -dimensional voxels can be interconnected exactly through common faces of dimension k or higher.

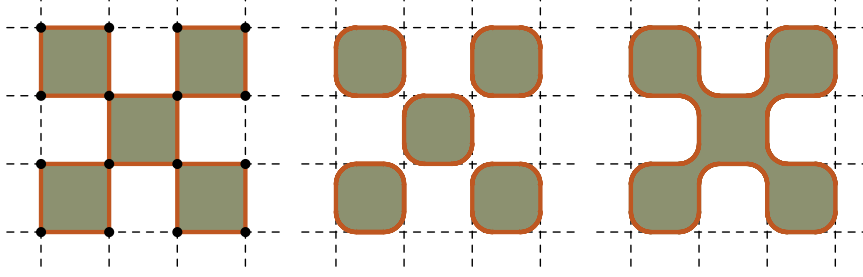


Figure 4: Fragment of a digital image and its topological implementation for the cases of 4- and 8-continuity.

Thus, to obtain $|K|$, we consider the disjointed union of all d -dimensional voxels from K and identify them by using common faces of dimension $\geq k$. Because two-dimensional (2D) digital images are considered in this article, we can discuss only two types of continuity, and these are usually denoted as 4- and 8-continuity (based on the maximum numbers of neighbors associated with a given pixel), as in figure 4.

We assume that the pore space has 4-continuity, and the matrix (skeleton) has 8-continuity. The pore space and the matrix complement each other, which is consistent with the duality for digital images, in which 4 and 8-continuities complement each other. There are three topological invariants in dimension two: the Betti numbers β_0 , β_1 and the Euler characteristic $\chi = \beta_0 - \beta_1$. Because of the duality indicated above, if $\bar{K}$ is a complement to K with dual continuity, $\beta_0(K) = \beta_1(\bar{K})$ and $\beta_1(K) = \beta_0(\bar{K}) - 1$. Thus, when calculating the topological characteristics of 2D images, it is possible to work with only pore space or only the skeleton. By contrast, all calculations can be reduced to the calculation of only the zero Betti numbers of the skeleton and the pore space. For calculating β_0 , we used an algorithm based on a disjoint-set data structure.

We now return to the construction of persistent homology in digital spaces and K a digital image. At this point, it is convenient to add to K all voxels that are faces of the d -dimensional voxels of $|K|$. Now, consider that the function $f : K \rightarrow \mathbb{R}$ is defined on the voxels of K and satisfies the following monotonic-

ity property: if a voxel σ contains the voxel τ as a face, $f(\sigma) \geq f(\tau)$. The monotonicity condition implies that the excursion sets K_a for all $a \in \mathbb{R}$ are CW subcomplexes of K . By running through all values of f , we obtain a *filtration* of the space K as follows:

$$K_0 = \emptyset \subset K_1 \subset \dots \subset K_n = K \quad (14)$$

(we denote $K_i = K_{a_i}$, where $a_1 < a_2 < \dots < a_n$ are values of function f).

For each pair of indices $0 \leq i < j \leq n$, the embedding $K_i \subset K_j$ induces a homomorphism of p -dimensional homology groups (we consider cellular homology with coefficients in $\mathbb{Z}_2$):

$$f_p^{i,j} : H_p(K_i) \rightarrow H_p(K_j). \quad (15)$$

The *p -dimensional persistent homology group* is the image of the homomorphism considered above: $H_p^{i,j} = \text{Im}(f_p^{i,j})$. The rank of this group $\beta_p^{i,j}$ is called the *p -dimensional persistent Betti number*. Let

$$\mu_p^{i,j} = (\beta_p^{i,j-1} - \beta_p^{i,j}) - (\beta_p^{i-1,j-1} - \beta_p^{i-1,j}), 0 \leq i < j \leq n. \quad (16)$$

It follows from (16) that $\mu_p^{i,j}$ equals the maximal number of independent p -dimensional homology classes that appear in K_i initially (that is, they are not images of classes from K_{i-1}) and disappear entering K_j . Consider the extended plane $\bar{\mathbb{R}}^2 = (\mathbb{R} \cup \{\pm\infty\})^2$ in which each point can have arbitrary multiplicity. The *p -dimensional persistence diagram* of function f is the set $PD_p(f)$ of points on the extended plane $\bar{\mathbb{R}}^2$ of the form (a_i, a_j) , $0 \leq i < j \leq n$, where each point takes the multiplicity $\mu_p^{i,j}$. We add to $PD_p(f)$ all points on the diagonal (a, a) , $a \in \mathbb{R}$ with infinite multiplicity. Figure 5 shows a 0-dimensional persistence diagram of the function f defined on a segment of $\mathbb{R}$.

The general algorithm for calculating persistent homologies is described in [17]. In our case, as described above, we must calculate only 0-dimensional persistence diagrams for 2D digital spaces, which allows us to simplify the general algorithm. The disjoint-set data structure fits very well with the filtration structure: moving along the subspaces $K_0, K_1, \dots$ or higher, we construct a system

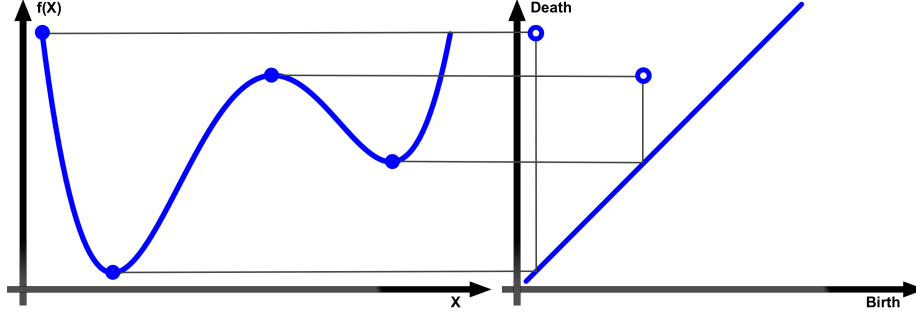


Figure 5: Persistence diagram of function defined on a segment.

of trees that describes the connected components of the next K_i by adding a pointer from each tree at the level $(i - 1)$ to the corresponding tree at the level i , which contains the previously connected component. By representing each connected component at each filtration level with a vertex and adding a pointer as a directed edge, we get a tree called the merge tree that contains all the information about the structure of the persistent zero-dimensional homology. By dividing the tree into separate edges according to the "Elder survives" rule [17], we obtain a persistence diagram in which each edge directed from level i to level j corresponds to a point with the coordinates (i, j) in the persistence diagram.

In practical applications, it is important to have a measure of the proximity between two persistence diagrams. Let PD_1 and PD_2 be two persistence diagrams. To determine the distance between them, we first consider the natural metric on the extended plane: we set the distance between two points $a = (a_1, a_2), b = (b_1, b_2) \in \mathbb{R}^2$ as equal to the L_∞ -norm:

$$\|a - b\|_\infty = \max\{|a_1 - b_1|, |a_2 - b_2|\}. \quad (17)$$

Furthermore, for any bijection $\varphi : PD_1 \rightarrow PD_2$, consider its diameter:

$$D(\varphi) = \sup_{a \in PD_1} \|a - \varphi(a)\|_\infty. \quad (18)$$

Then, we define the "bottleneck distance" between the diagrams PD_1, PD_2

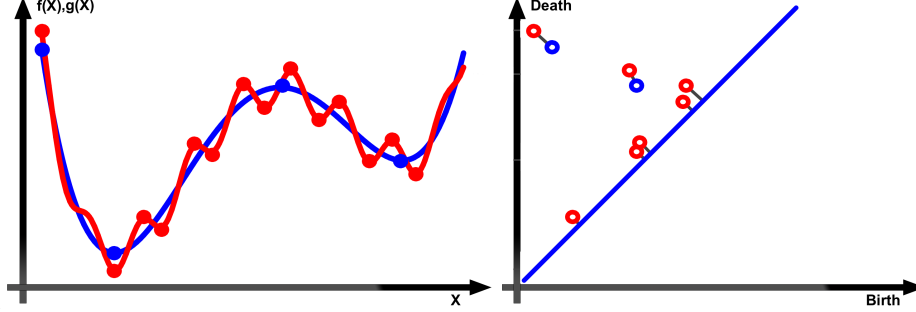


Figure 6: Stability of persistence diagrams with respect to perturbations in the L_∞ norm.

as follows:

$$W_\infty(PD_1, PD_2) = \inf_{\varphi: PD_1 \rightarrow PD_2} D(\varphi), \quad (19)$$

where the infimum is taken over all bijections φ . Note that bijections always exist because persistence diagrams contain diagonal points with infinite multiplicity. In addition, the closer the point of the persistence diagram to the diagonal, the smaller is its contribution to the distance to another diagram.

The stability of persistence diagrams with respect to perturbations of the function f in the norm L_∞ was proved in [27] (see an illustration of this phenomenon in figure 6). That is, let K be a digital image and $f, g : K \rightarrow \mathbb{R}$ be two monotone functions. Then, for every dimension p , we have the inequality:

$$W_\infty(PD(f), PD(g)) \leq \|f - g\|_\infty. \quad (20)$$

This result justifies the use of persistence diagrams for the topological analysis of our experimental results.

4. Numerical experiments

4.1. Fixed pore space

The first series of experiments was conducted for a fixed starting model of the pore space, as shown in figure 7. The model was constructed using the truncated Gaussian field method [28] with a correlation length of $5 \cdot 10^{-5}$ m and 67% porosity. Such a high level of porosity is nonphysical in three-dimensional

(3D) case; however, in the 2D case, it ensures percolation of the samples. We set the grid step to 10^{-5} m. Thus, the correlation length of the random field was five grid steps.

We considered the rock matrix as monomineralic calcite with a mass density equaling $\rho = 2710$ kg/m³ and stoichiometric coefficient equaling 1. The fluid was CO_2 -saturated brine with an equilibrium pH of 7, and the pH at the inlet was 3.1 (as provided in [6] page 108). We assumed the physical properties of the fluid to be independent of its pH. Therefore, we fixed the dynamic viscosity as $\mu = 0.00028$ Pa·s, which corresponds to water at 100 degrees centigrade (at the formation conditions). In the first series of experiments, we varied three parameters, namely, pressure drop between the inlet and outlet denoted as $dP \in [5 \cdot 10^{-2}; 5]$ Pa, diffusion coefficient $D \in [9.3 \cdot 10^{-9}, 9.3 \cdot 10^{-7}]$, and reaction rate $k_r \in [1.5 \cdot 10^{-4}, 1.5 \cdot 10^{-2}]$. These three parameters were independently and homogeneously distributed within their intervals of variability. The simulation time was 300 s. To illustrate the results of the simulation, in figure 8; we show the solution pH ($pH = -\lg C$, where C is the concentration of H^+ cations) at $t = 300$ s across four experiments.

We performed 475 simulations with different triplets (dP, D, k_r) . Each simulation was supported by computation of the persistence diagrams for the connected components of the matrix. The complete datasets of the simulation results can be found in [29]. Thereafter, we computed the pairwise distances between the diagrams and applied the k-means clustering technique with respect to our metric. We defined four different clusters and grouped the experiments accordingly. The distance matrix for the regrouped experiments is shown in figure 9. Note that two of the four clusters are distinguishable, while the other two are relatively close.

To analyze the effects of the input parameters on the topology changes, we plotted the points in 3D space (dP, D, k_{chem}) in figure 10, where each point corresponds to an individual experiment. The color of the marker represents the cluster to which an experiment belongs. In this section, we divide these experiments into four clusters and we fix the correspondence between the cluster

number and its color: blue - first, red - second, green - third, and black - fourth.

It is reasonable to start the analysis with the most significant cluster, that is, the second or red cluster because it contains more than half of all the experiments. The input parameter triplets (dP, D, k_{chem}) for the experiments belonging to this cluster correspond to two completely different cases. In the first case, the reaction rate is low, and the other two parameters can be arbitrary. Consider the Damkohler number $Da = \frac{lk_r}{D}$, where l is a characteristic length scale equal to the correlation length in this research. This case corresponds to a low $Da \in [7.5 \cdot 10^{-3}, 7.5 \cdot 10^{-1}]$, and the reaction is diffusion-limited. In the second case, the diffusion coefficient is low, and the reaction rate and pressure drop can be arbitrary. Thus, $Da \in [7.5 \cdot 10^{-1}, 7.5 \cdot 10^3]$ is relatively high, which causes the reaction to be kinetic-limited. Consequently, the solution behaves differently. In both cases, the reactant can fill the pore space owing to convection (figure 8, cluster 2), but almost no changes occur in the pore space because the reactant has limited access to the interface, as in the first case, or because of the low reaction rate, as in the second case. In both cases, changes in the topology of the pore space or the rock matrix are indistinguishable in terms of the persistence diagram metric.

The increases in both the diffusion coefficient and the reaction rate lead to different types of topology changes. The green cluster corresponds to intermediate values of the diffusion and reaction rates. In this case, transport of the active component (due to diffusion) is sufficient to support the reaction. However, the reaction rate is too high to permit convection-governed reactant transport. Consequently, a relatively fast reaction occurs near the inlet boundary, leading to the formation of a "reaction front" (figure 8, cluster 3).

For the blue cluster, the diffusion coefficient further increases, which improves the access of the reactant to the reacting surfaces under the same reaction rate as that in the green cluster. Therefore, the reactant penetrates deeper into the sample, resulting in the formation of conical wormholes in the sample (figure 8, cluster 1). The last (black) cluster is the smallest, albeit the most distant cluster. It corresponds to the highest values of k_r and D , thus providing

the best support to the reaction through diffusion of the reactant toward the interface. This results in the formation of elongated wormholes in the sample (figure 8, cluster 4).

In addition to a formal analysis of the topology changes, we considered the evolution of the transport properties of the sample and their dependence on different types of pore structure changes. We focused on the estimation of porosity, permeability, and tortuosity [20] because these parameters are used in macroscopic fluid-transport models [30] and for determining wave propagation in poroelastic media [31], [32].

The porosity is estimated as

$$\psi = \frac{N^+}{N},$$

where N^+ is the number of grid cells in which the level-set is positive, and N is the total number of grid cells in the model. The permeability (horizontal component of the permeability tensor) is estimated using the following relationship:

$$\kappa^{-1}(x_1) \frac{Q}{S} = -\frac{\partial P}{\partial x_1},$$

where flux $Q(x_1) = \int_0^{L_1} u_1 dx_2$, and $S = L_2$ is the cross-sectional area. Here, we assume $u_1(x_1, x_2) = 0$ if $\varphi(x_1, x_2) \leq 0$. The flux Q is supposed to be independent of x_1 ; thus, the permeability averaged over the sample length $x \in [0, L_1]$ can be expressed as

$$\hat{\kappa}^{-1} \frac{Q}{S} = -\frac{dP}{L_1}.$$

To reduce the effect of the numerical errors, we use the mean value of flux over x_1 . Thus, the final expression for estimating permeability is

$$\hat{\kappa} = -\frac{\sum_{i,j} (u_1)_{i+1/2,j} h_2}{N_1} \frac{h_2}{dP}.$$

To estimate the hydraulic tortuosity along the x_1 direction, we follow the definition given in [33], where the tortuosity is computed as the ratio of the mean absolute value of the velocity vector over the mean of the tangential component

of the velocity vector (formula 6 in [33])

$$T = \frac{\int |\vec{u}(x_1, x_2)| dx_1 dx_2}{\int u_1(x_1, x_2) dx_1 dx_2} = \frac{\sum_{i,j} \sqrt{((u_1)_{i+1/2,j} + (u_1)_{i-1/2,j})^2 + ((u_2)_{i,j+1/2} + (u_2)_{i,j-1/2})^2}}{\sum_{i,j} ((u_1)_{i+1/2,j} + (u_1)_{i-1/2,j})}.$$

We interpolate the values of the velocity vector to the centers of the grid cells, because originally they are defined at different grid points in the staggered grid.

We consider the set of output parameters $(\varphi(t), \hat{\kappa}(t), T(t))$ as a 3D curve of one parameter. This parameter is model time. In figures 11 and 12, we show 2D projections of these curves onto the planes $(\varphi, \hat{\kappa})$ and (φ, T) respectively. The thin lines represent an individual experiment, whereas the thick lines correspond to the "mean experiment." Note that the "mean experiment" is defined in the bottleneck metric of the persistence diagrams, and it is not the same as the average of the thin lines. If porosity-tortuosity relationship is considered, for the second (red) cluster, tortuosity decreases slowly as porosity increases. This effect can be ascribed to the low rate of matrix dissolution without the channel formation. In other cases, wormhole formation leads to a noticeable decrease in tortuosity as porosity increases, which leads to an increase in permeability as well (figure 11).

4.2. Statistical realizations of the models

In the second series of experiments, we varied the pore space models. As before, we constructed the models by using the truncated Gaussian field method [28] but for each experiment, we used a new realization. The correlation length of the Gaussian field was $5 \cdot 10^{-5}$ m, and porosity was 67%, same as that in the previous section. However, we considered the triplet (dP, D, k_{chem}) in a regular grid. The parameter values are listed in table 2. For each triplet value, we considered up to 10 realizations of the model. In total, we performed 579 experiments. All other model parameters, such mass density of the matrix and fluid viscosity, were the same as those in the previous experiments. However, to enhance the topology changes, we employed a higher reactant concentration, and consequently, $pH = 2.3$ at the inlet.

The high concentration of cations at the inlet and wide range of reaction rates led to a rich set of diverse geometries and topologies of the pore space, varying from full matrix dissolution to no reaction at all. In the two limiting cases, namely fast dissolution and no dissolution, almost no topology changes were observed over the major period. Therefore, the distance between the two persistence diagrams was small, even though they corresponded to completely different physical processes. In the intermediate cases, the persistence diagrams were richer because several new connectivity components appeared and disappeared over the considered time. We show the persistence diagrams for six representative experiments (figure 13). The input parameters for the simulations are listed in table 1. Note that if the dissolution is fast, all connectivity components appear and disappear within a short period, for instance, in 20 time steps in experiment 172 (black markers), 40 steps in experiment 108 (red markers), and 80 steps in experiment 115 (green markers). In experiment 143 (pink markers), dissolution was sustained over the entire experimental period. Consequently, the new components appeared at high time instants. It is worth considering experiments 26 (blue) and 11 (light-blue) together. In both cases, the persistence diagrams were poor. However, in experiment 11 (light-blue), only one connectivity component disappeared during the experiment; thus, almost no dissolution occurred. By contrast, in experiment 26, a few components disappeared, which may indicate slow dissolution near the inlet.

Table 1: Input parameters for typical experiments

exp.	11	26	108	115	143	172
p_{bc}	$5 \cdot 10^{-2}$	$5 \cdot 10^{-2}$	$5 \cdot 10^{-1}$	$5 \cdot 10^{-1}$	$5 \cdot 10^0$	$5 \cdot 10^0$
D	$9.3 \cdot 10^{-9}$	$9.3 \cdot 10^{-9}$	$9.3 \cdot 10^{-9}$	$9.3 \cdot 10^{-9}$	$9.3 \cdot 10^{-9}$	$9.3 \cdot 10^{-9}$
k_{chem}	$1.5 \cdot 10^{-4}$	$1.5 \cdot 10^{-3}$	$1.5 \cdot 10^{-1}$	$1.5 \cdot 10^{-1}$	$1.5 \cdot 10^{-3}$	$1.5 \cdot 10^{-1}$

However, if k-means clustering were to be applied to the persistence diagrams to divide all 579 experiments into two clusters, one cluster would contain experiments 26, 108, and 172. The other cluster would contain experiments

115, 143, and 11. This means that the persistence diagram metric correlates well with the dissolution rate within the specified limits. However, for the case of extremely rapid and prolonged dissolution, changes in the persistence diagrams are negligible and cannot be distinguished. This effect can be observed even when the number of clusters is increased. In figure 14, we show plots of the experimental input parameters in the 3D space of the triplet (dP, D, k_r) , where different colors correspond to different clusters. Moreover, in figure 15 we show the dependence of porosity on time for division into two and four clusters. As in the previous section, the thin lines correspond to individual experiments and the thick lines to the "mean experiment" in the bottleneck metric of the persistence diagrams, that is, topologically mean experiment over a cluster. In both cases, we could stably recover the clusters associated with a high reaction rate (blue denotes division into two clusters, and red denotes division into four clusters). In these experiments, the dissolution rate remained high until the matrix was completely dissolved. However, in both cases, the discussed clusters included the experiments with low diffusion coefficients as well, which correspond to no dissolution. All intermediate situations were assigned to the other clusters. In particular, in the case of only two clusters, the experiments in which the dissolution rates were moderate were assigned to the red cluster. In the case of four clusters, these experiments were assigned to the blue, green, and black clusters depending on the dissolution rate. Thus, the persistence diagrams of the matrix connectivity components correlate to the dissolution rate and can be used to estimate it. However, the diagrams were not useful for distinguishing between the limiting cases of rapid and prolonged dissolution rates, because in both cases, in general, the topologies of the pore space and the matrix remained constant.

In addition, the considered sample size of 250 by 250 points or correlation length of 50 by 50 is representative for fluid flow and topology changes. First, we measured the permeability and tortuosity of the initial models, that is, before the simulation of matrix dissolution, which corresponds to $t = 0$ s. Their dispersion over different realizations did not exceed 10 %, which fulfills the definition of the

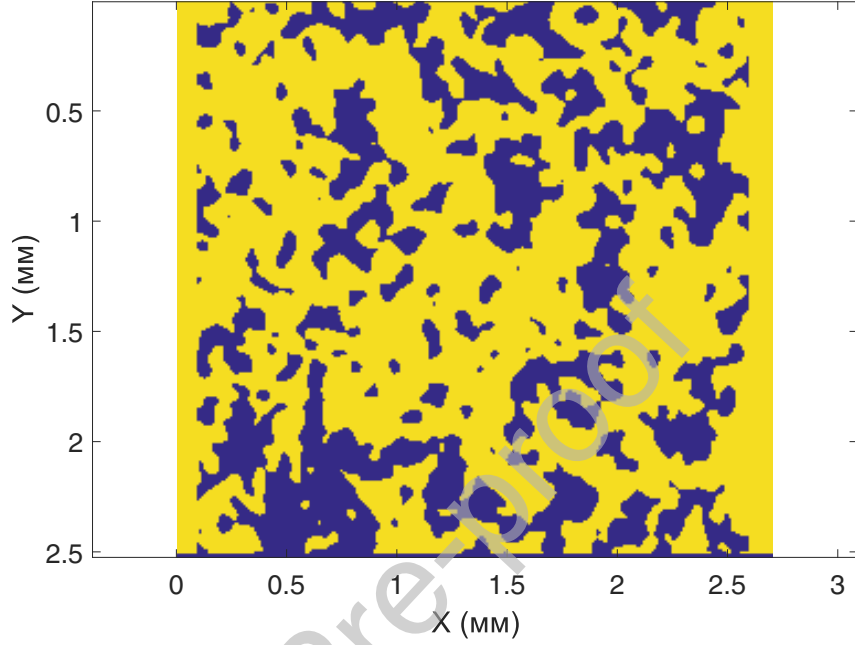


Figure 7: Pore space model used for the first series of numerical experiments. The light regions represent the pore space, and the dark regions correspond to the matrix.

statistically representative volume [34]. Second, clustering stably divided the experiments according to the input parameters, where experiments conducted using different model realizations but with the same input parameters typically fell into the same cluster.

Table 2: Input parameters

p_{bc}	$5 \cdot 10^{-2}$	$5 \cdot 10^{-1}$	$5 \cdot 10^0$	
D	$9.3 \cdot 10^{-9}$	$9.3 \cdot 10^{-8}$	$9.3 \cdot 10^{-7}$	
k_{chem}	$1.5 \cdot 10^{-4}$	$1.5 \cdot 10^{-3}$	$1.5 \cdot 10^{-2}$	$1.5 \cdot 10^{-1}$

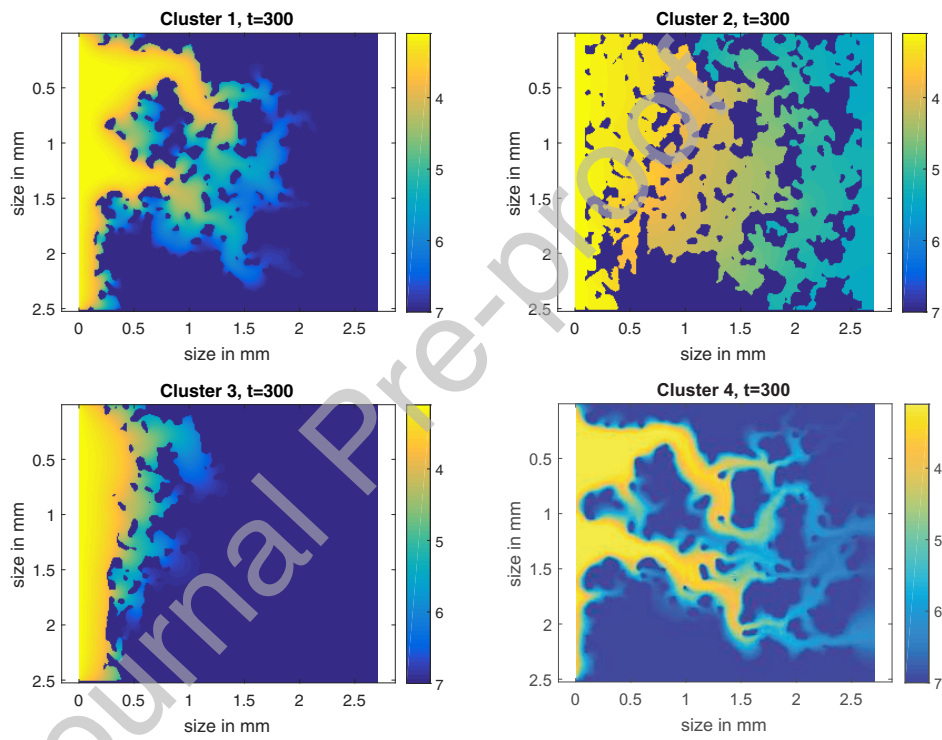


Figure 8: Solution pH at $t = 300$ s for different combinations of the triplet (dP, D, k_r) . The models correspond to different types of topology evolutions.

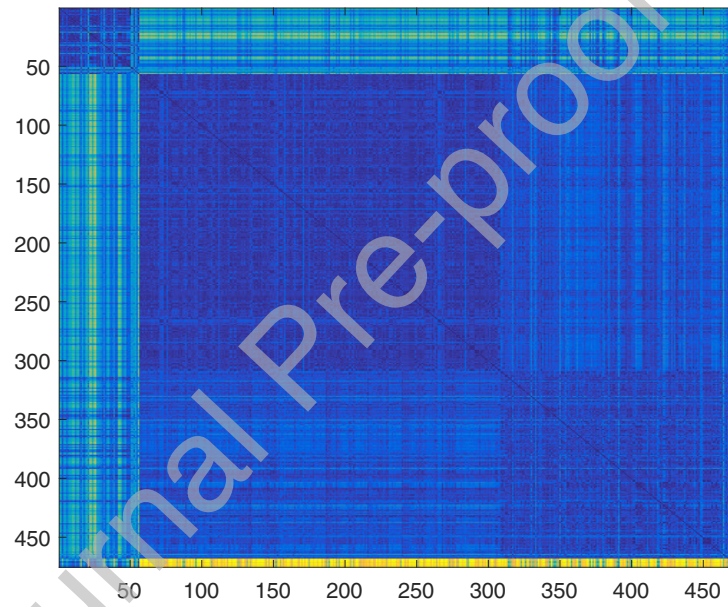


Figure 9: Distance matrix for the clustered persistence diagrams.

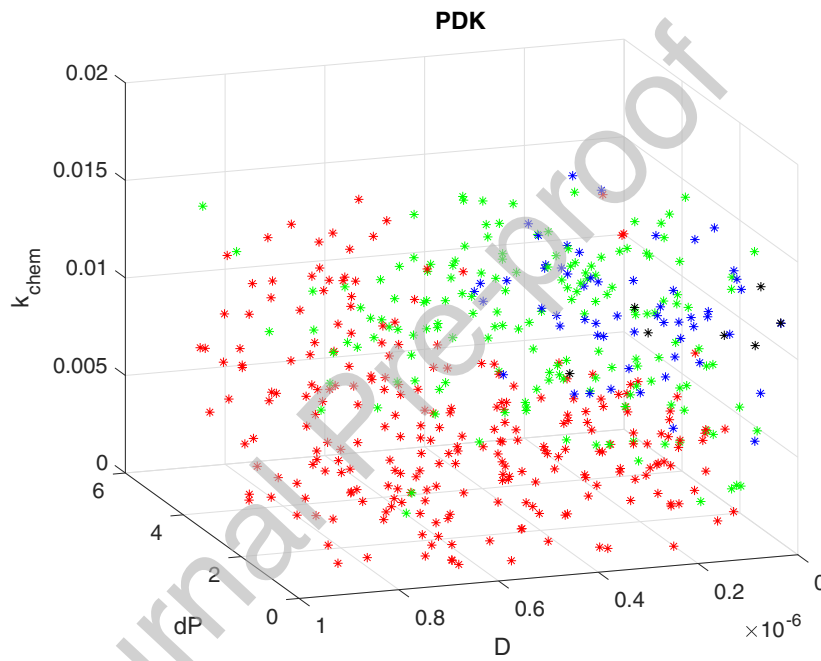


Figure 10: Correspondence between the input parameters (dP, D, k_{chem}) and the clustering with respect to persistence diagrams. The positions of the points correspond to the coordinates in the space of the input parameters, and the colors correspond to the different clusters.

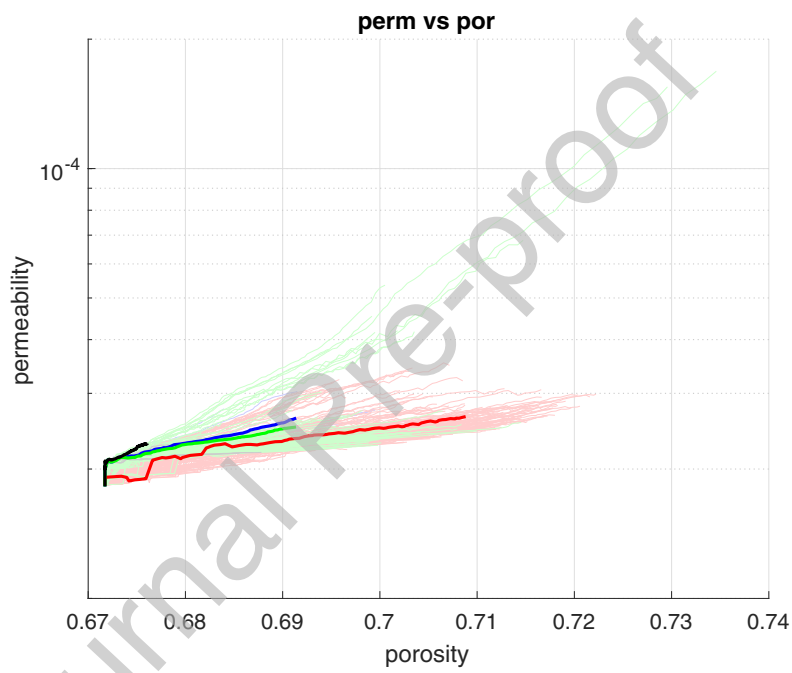


Figure 11: Porosity-permeability curves. The thin lines correspond to individual experiments, thick lines to the mean values (in terms of the persistence diagram metric). Different colors represent different clusters.

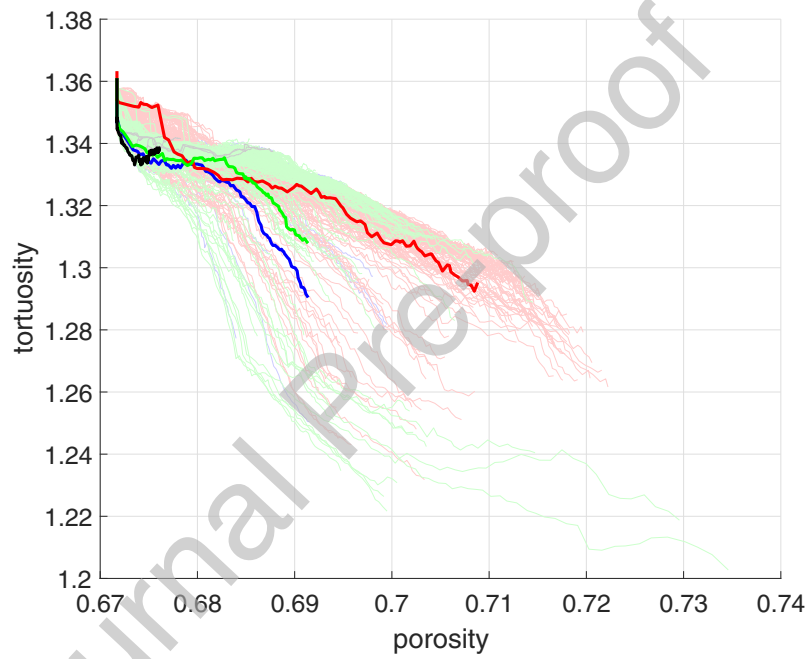


Figure 12: Porosity-tortuosity curves. The thin lines correspond to individual experiments, thick lines to the mean values (in terms of the persistence diagram metric). Different colors represent different clusters.

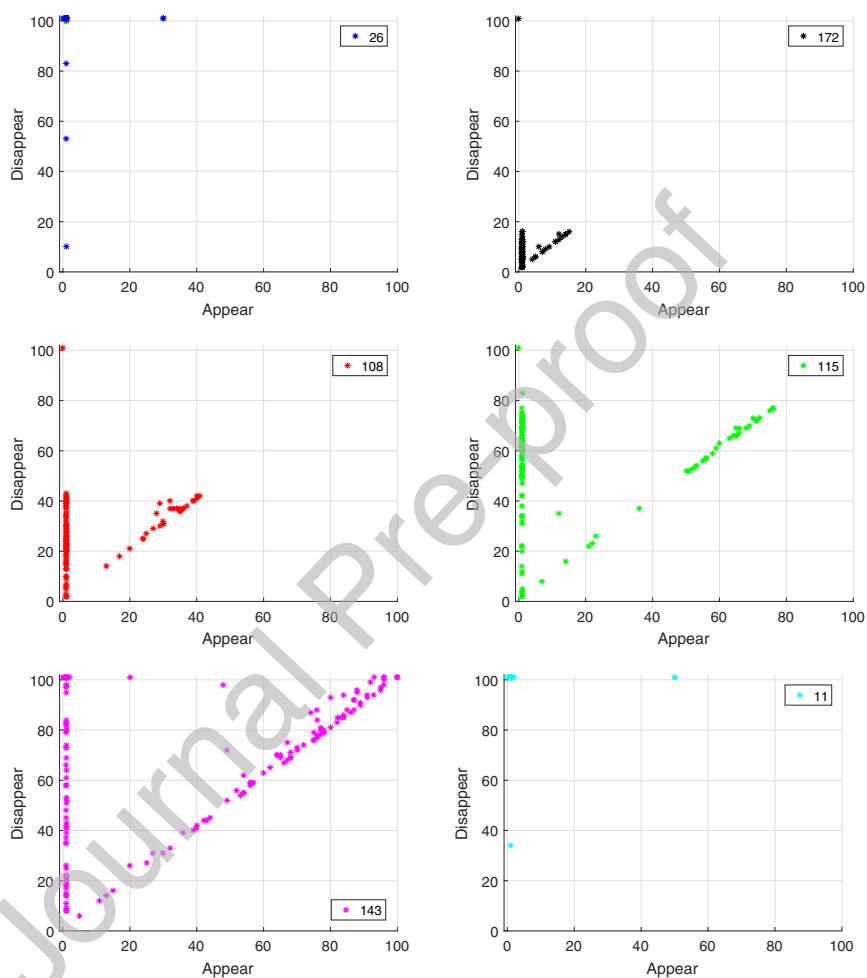


Figure 13: Persistence diagrams for six different experiments. In experiments 26 (blue markers) and 11 (light-blue markers), the rock matrix did not change. In experiments 172 (black), 108 (red), 115 (green), and 143 (pink), the dissolution rate decreased from experiment 172 to 143.

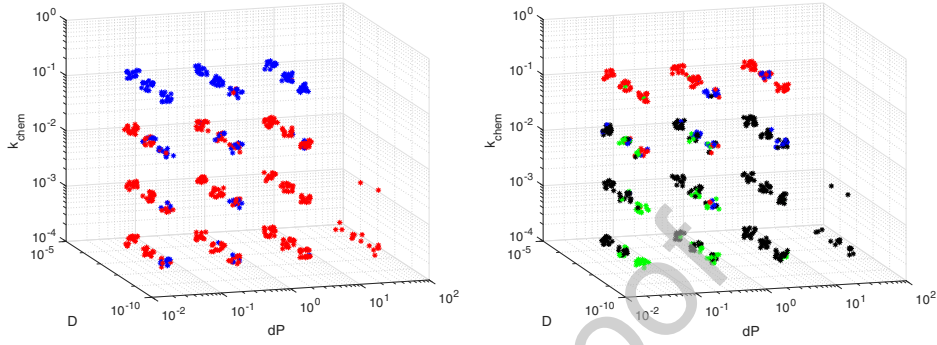


Figure 14: Correspondence between the input parameters (dP, D, k_{chem}) and the clustering with respect to the persistence diagrams. The positions of the points correspond to the coordinates in the space of the input parameters, and the colors correspond to the clusters. The plot on the left represents division into two clusters, and the plot on the right corresponds to division into four clusters.

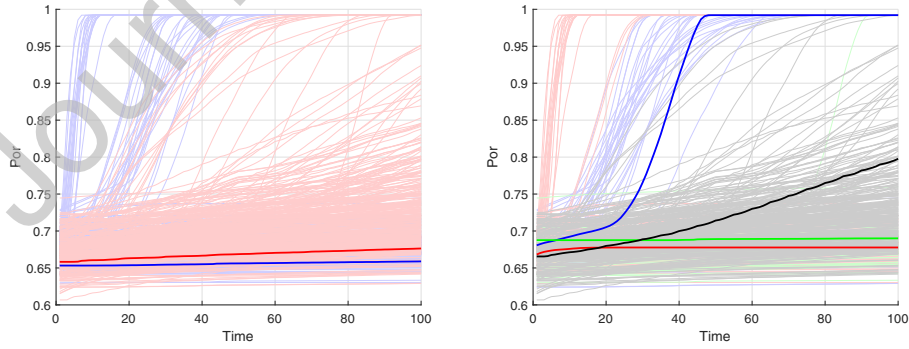


Figure 15: Evolution of porosity with time for two clusters (left) and four clusters (right).

5. Conclusions

We presented an algorithm for numerical simulation of reactive transport at the pore scale. The algorithm was based on the level-set method, which can handle changes in pore space and matrix topology. Simulation of reactive transport at the pore scale allowed us to observe pore space evolution and estimate hydrodynamic parameters of the core samples, such as permeability and hydraulic tortuosity. Thus, the algorithm is oriented toward the numerical upscaling and reconstruction of the coefficients in the equations of state, which are used in reactive transport models at the reservoir scale, for example, porosity-permeability-tortuosity relationships.

Moreover, we suggested a new methodology to quantitatively classify the pore space evolution during chemical fluid-solid interaction. The approach was based on persistence homology, that is, the appearance and disappearance (opening of the isolated pore) times of each connectivity component. The use of the bottleneck metric in the space of the persistence diagrams allowed us to apply clustering techniques, such as k-means, to distinguish different scenarios of core matrix dissolution. Specifically, in the case of low chemical activity of the fluid, topological clustering stably separated the cases of homogeneous dissolution of the core matrix, formation of wormholes, and formation of a dissolution front near the inlet. In the case of high chemical activity, the persistence diagrams indicated the rate of the matrix dissolution. Moreover, the clustering of different dissolution scenarios on the basis of the persistence homology was found to be correlated to differences in the cross-property relationships (porosity-permeability-tortuosity), which can be used in reservoir-scale reactive transport simulations.

Note that construction of the persistence diagrams and their classification was very computationally efficient, requiring less than 1 % of the computational time, which is essential for reactive transport simulation. Therefore, the proposed approach can be extended to 3D to compute the diagrams of either connectivity components (pores) or independent cycles (analogue of throats in

pore-network models) for efficient characterization of the pore space and core-matrix evolution during chemical fluidsolid interaction

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References

- [1] M. Ghommem, W. Zhao, S. Dyer, X. Qiu, D. Brady, Carbonate acidizing: Modeling, analysis, and characterization of wormhole formation and propagation, *Journal of Petroleum Science and Engineering* 131 (2015) 18–33.
- [2] K. Miller, T. Vanorio, Y. Keehm, Evolution of permeability and microstructure of tight carbonates due to numerical simulation of calcite dissolution, *Journal of Geophysical Research: Solid Earth* 122 (6) (2017) 4460–4474.
- [3] A. Dadda, C. Geindreau, F. Emeriault, S. R. du Roscoat, A. Garandet, L. Sapin, A. E. Filet, Characterization of microstructural and physical properties changes in biocemented sand using 3d x-ray microtomography, *Acta Geotechnica* 12 (5) (2017) 955–970.
- [4] A. H. Alizadeh, M. Akbarabadi, E. Barsotti, M. Piri, N. Fishman, N. Nagarajan, Salt precipitation in ultratight porous media and its impact on

- pore connectivity and hydraulic conductivity, *Water Resources Research* 54 (4) (2018) 2768–2780.
- [5] E. Kaya, S. J. Zarrouk, Reinjection of greenhouse gases into geothermal reservoirs, *International Journal of Greenhouse Gas Control* 67 (2017) 111–129.
 - [6] M. Lebedev, Y. Zhang, M. Sarmadivaleh, A. Barifcani, E. Al-Khdheawi, S. Iglaier, Carbon geosequestration in limestone: Pore-scale dissolution and geomechanical weakening, *International Journal of Greenhouse Gas Control* 66 (2017) 106–119.
 - [7] C. I. Steefel, C. A. J. Appelo, B. Arora, D. Jacques, T. Kalbacher, O. Kolditz, V. Lagneau, P. C. Lichtner, K. U. Mayer, J. C. L. Meeussen, S. Molins, D. Moulton, H. Shao, J. imnek, N. Spycher, S. B. Yabusaki, G. T. Yeh, Reactive transport codes for subsurface environmental simulation, *Computational Geosciences* 19 (3) (2015) 445–478.
 - [8] T. B. Costa, K. Kennedy, M. Peszynska, Hybrid three-scale model for evolving pore-scale geometries, *Computational Geosciences* 22 (3) (2018) 925–950.
 - [9] S. Molins, D. Trebotich, L. Yang, J. B. Ajo-Franklin, T. J. Ligocki, C. Shen, C. I. Steefel, Pore-scale controls on calcite dissolution rates from flow-through laboratory and numerical experiments, *Environmental Science and Technology* 48 (13) (2014) 7453–7460.
 - [10] G. Yang, Y. Li, A. Atrens, D. Liu, Y. Wang, L. Jia, Y. Lu, Reactive transport modeling of long-term co2 sequestration mechanisms at the shenhua ccs demonstration project, china, *Journal of Earth Science* 28 (3) (2017) 457–472.
 - [11] Q. Kang, L. Chen, A. J. Valocchi, H. S. Viswanathan, Pore-scale study of dissolution-induced changes in permeability and porosity of porous media, *Journal of Hydrology* 517 (2014) 1049–1055.

- [12] H. Yoon, A. J. Valocchi, C. J. Werth, T. Dewers, Pore-scale simulation of mixing-induced calcium carbonate precipitation and dissolution in a microfluidic pore network, *Water Resources Research* 48 (2) (2012) W02524.
- [13] S. Osher, R. P. Fedkiw, Level set methods: An overview and some recent results, *Journal of Computational Physics* 169 (2) (2001) 463–502.
- [14] Z. Xu, P. Meakin, Phase-field modeling of solute precipitation and dissolution, *The Journal of Chemical Physics* 129 (1) (2008) 014705.
- [15] C. S. Peskin, Flow patterns around heart valves: A numerical method, *Journal of Computational Physics* 10 (1972) 252–271.
- [16] F. Sotiropoulos, X. Yang, Immersed boundary methods for simulating fluidstructure interaction, *Progress in Aerospace Sciences* 65 (2014) 1–21.
- [17] H. Edelsbrunner, J. Harer, *Computational Topology, An Introduction*, Amer. Math. Soc., 2010.
- [18] T. Y. Kong, A. Rosenfeld, Digital topology: Introduction and survey, *Computer Vision, Graphics, and Image Processing* 48 (3) (1989) 357–393.
- [19] K. Turner, Y. Mileyko, S. Mukherjee, J. Harer, Frechet means for distributions of persistence diagrams, *Discrete and Computational Geometry* 52 (1) (2014) 44–70.
- [20] Y. Bazaikin, B. Gurevich, S. Iglaier, T. Khachkova, D. Kolyukhin, M. Lebedev, V. Lisitsa, G. Reshetova, Effect of ct image size and resolution on the accuracy of rock property estimates, *Journal of Geophysical Research: Solid Earth* 122 (5) (2017) 3635–3647.
- [21] F. Bouchelaghem, R. Pusch, Fluid flow and effective conductivity calculations on numerical images of bentonitemicrostructure, *Applied Clay Science* 144 (2017) 9–18.
- [22] K. M. Gerke, M. V. Karsanina, R. Katsman, Calculation of tensorial flow properties on pore level: Exploring the influence of boundary conditions on

- the permeability of three-dimensional stochastic reconstructions, *Physical Review E* 100 (5) (2019) 053312.
- [23] F. Gibou, R. Fedkiw, S. Osher, A review of level-set methods and some recent applications, *Journal of Computational Physics* 353 (2018) 82–109.
- [24] H. Andra, N. Combaret, J. Dvorkin, E. Glatt, J. Han, M. Kabel, Y. Keehm, F. Krzikalla, M. Lee, C. Madonna, M. Marsh, T. Mukerji, E. H. Saenger, R. Sain, N. Saxena, S. Ricker, A. Wiegmann, X. Zhan, Digital rock physics benchmarks - part i: Imaging and segmentation, *Computers and Geosciences* 50 (0) (2013) 25–32.
- [25] R. Mittal, G. Iaccarino, Immersed boundary methods, *Annual Review of Fluid Mechanics* 37 (1) (2005) 239–261.
- [26] R. P. Fedkiw, T. Aslam, B. Merriman, S. Osher, A non-oscillatory eulerian approach to interfaces in multimaterial flows (the ghost fluid method), *Journal of Computational Physics* 152 (2) (1999) 457–492.
- [27] D. Cohen-Steiner, H. H. Edelsbrunner, Stability of persistence diagrams, *J. Discrete Comput Geom.* 37 (2007) 103–120.
- [28] J. D. Hyman, C. L. Winter, Stochastic generation of explicit pore structures by thresholding gaussian random fields, *Journal of Computational Physics* 277 (2014) 16–31.
- [29] V. Lisitsa, Y. Bazaikin, T. Khachkova, Barcodes of rock matrix during chemical dissolution, *Mendeley Data* (2020). doi:10.17632/9jndrnghj5.1.
- [30] A. E. Amikiya, M. K. Banda, Modelling and simulation of reactive transport phenomena, *Journal of Computational Science* 28 (2018) 155–167.
- [31] M. A. Biot, Theory of propagation of elastic waves in fluid-saturated porous solid. i. low-frequency range, *Journal of the Acoustical Society of America* 28 (1956) 168–178.

- [32] E. Caspari, M. Novikov, V. Lisitsa, N. D. Barbosa, B. Quintal, J. G. Rubino, K. Holliger, Attenuation mechanisms in fractured fluid-saturated porous rocks: a numerical modelling study, *Geophysical Prospecting* 67 (4) (2019) 935–955.
- [33] A. Duda, Z. Koza, M. Matyka, Hydraulic tortuosity in arbitrary porous media flow, *Physical Review E* 84 (3) (2011) 036319, pRE.
- [34] D. Zhang, R. Zhang, S. Chen, W. E. Soll, Pore scale study of flow in porous media: Scale dependency, rev, and statistical rev, *Geophysical Research Letters* 27 (8) (2000) 1195–1198.